

TECHBIZ SECURITY

Cyber Security Internship Program

Windows Security Event Log Investigation and SOC Analysis

Week 2 - Task 5 (SOC)

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1. Introduction

This report documents the work completed during Week 2, Task 5 (SOC) of the TechBiz Security Cyber Security Internship. The task focused on developing practical Security Operations Center (SOC) investigation skills through the review of Windows Security Event Logs on a lab Windows Server environment, and through the completion of related TryHackMe learning activities. The exercise required reviewing key Windows Security Event IDs, understanding their significance to a SOC analyst, and practicing the correlation of authentication, process-creation and account-management events during an investigation.

All findings presented in this report are based strictly on the Event Viewer screenshots and TryHackMe completion screenshots captured during the task. Where a specific detail could not be clearly read from the available evidence, this is explicitly noted rather than assumed.

2. Internship Task Overview

Task 5 of Week 2 required the review of Windows Security Event Logs on a Windows Server 2022 lab machine (hostname WIN-BCDCP359AAS) using the built-in Event Viewer application. The investigation covered four specific Windows Security Event IDs: 4624 (successful logon), 4625 (failed logon), 4688 (process creation) and 4720 (user account creation). In addition, the task required completion of the TryHackMe “Investigating Windows” room and the TryHackMe “Windows Fundamentals 1” room to reinforce practical Windows forensic and fundamentals knowledge.

3. Task Objectives

- Review Windows Security Event Logs using Event Viewer on a lab Windows Server machine.
- Identify and analyze Event ID 4624 (successful account logon) entries.
- Identify and analyze Event ID 4625 (failed account logon) entries, including detailed event properties.
- Identify and analyze Event ID 4688 (process creation) entries.
- Identify and analyze Event ID 4720 (user account creation) entries.
- Complete the TryHackMe “Investigating Windows” room to practice Windows investigation and forensic analysis concepts.
- Complete the TryHackMe “Windows Fundamentals 1” room to reinforce core Windows operating system concepts.
- Apply basic SOC investigation methodology, including event correlation and timeline analysis.

4. SOC Investigation Environment

The investigation was performed on a Windows Server lab machine running inside an Oracle VirtualBox virtual machine, named “WinServer2022-Agent.” The Windows Security event log on this host contained 24,842 total events at the time of review. The Event Viewer console was used to filter the Security log by specific Event ID values to isolate the events relevant to this investigation. Table 1 summarizes the environment details visible in the evidence.

Attribute	Observed Value
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Virtual Machine Name	WinServer2022-Agent
Hostname / Computer Name	WIN-BCDCP359AAS
Log Reviewed	Security
Total Events in Security Log	24,842
Event Source	Microsoft Windows security auditing
Tool Used	Windows Event Viewer

5. Windows Event Logging Overview

Windows Security Event Logs record security-relevant activity on a Windows host, including authentication attempts, process execution and account management actions. Each event is identified by a numeric Event ID, a Task Category and a Keyword indicating success (Audit Success) or failure (Audit Failure). For a SOC analyst, these logs are a primary source of telemetry for detecting suspicious authentication activity, unauthorized process execution and unexpected account changes. Table 2 summarizes the four Event IDs examined during this task.

Event ID	Event Type	SOC Relevance
4624	Successful Logon	Confirms successful authentication; used to verify legitimate access and to establish logon timelines.
4625	Failed Logon	Indicates failed authentication attempts; monitored for repeated failures or unusual login patterns.
4688	Process Creation	Records new process creation; used to correlate execution activity with potentially suspicious behavior.
4720	User Account Creation	Indicates a new user account was created; monitored for unexpected or unauthorized account creation.

6. Investigation of Event ID 4624 – Successful Logon

Event ID 4624 represents a successful account logon on a Windows system and is generated whenever a logon session is created. Figure 1 shows the Event Viewer Security log filtered to Event ID 4624, which returned 4,981 matching events out of the 24,842 total Security log entries on the host.

Observed Evidence

The Event Viewer shows a filtered view of the Security log (Log Name: Security) with the Source listed as Microsoft Windows security auditing, Task Category “Logon,” and Keyword “Audit Success.” The selected event in the screenshot has the following properties: Event ID 4624, Level Information, Logged 29-08-2026 2:48:54 PM, Computer WIN-BCDCP359AAS, and the general description “An account was successfully logged on.” The User field is listed as N/A in the General tab. Multiple 4624 events are visible in the list, all marked Audit Success under the Logon task category.

SOC Relevance

Successful logon events are important baseline telemetry for a SOC analyst. While a 4624 event on its own is not evidence of malicious activity, it is used to confirm when and how an authenticated session began, and it becomes significant when correlated with other events — for example, when a successful logon follows a series of failed logon attempts, or when a successful logon occurs from an unexpected account, at an unusual time, or immediately before suspicious process activity. Analysts also examine the Logon Type field (not fully visible in this screenshot) to distinguish between interactive, network, remote desktop and service logons, as this context shapes the investigation.

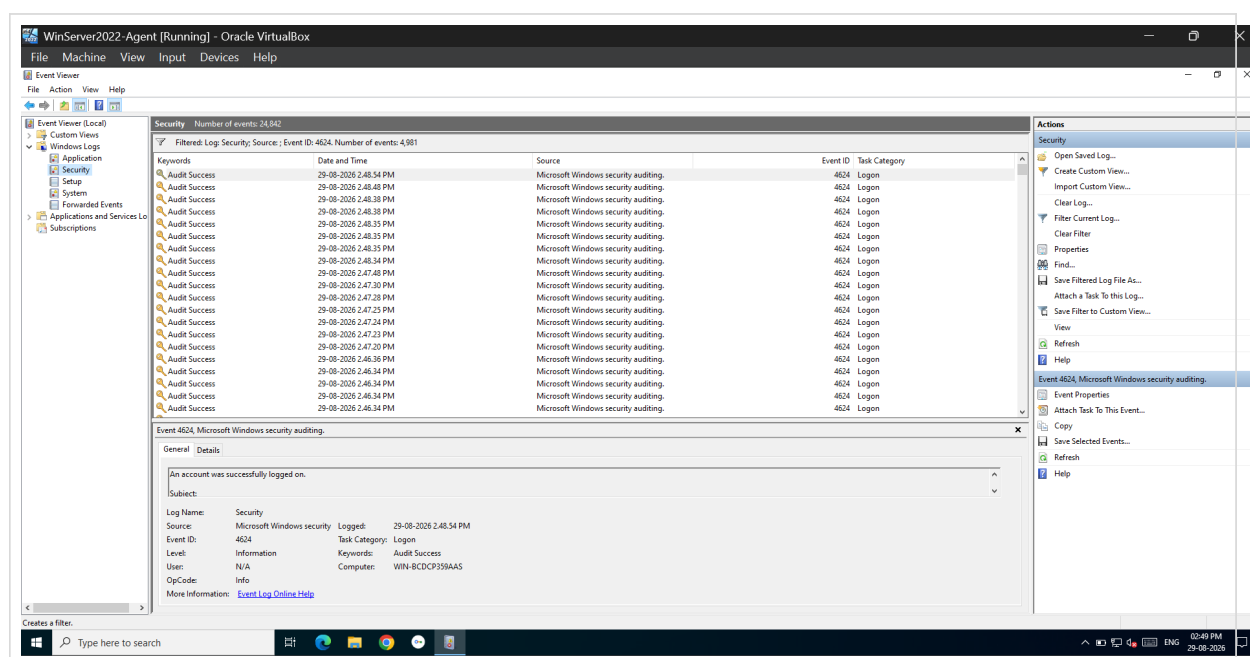


Figure 1: Windows Security Event ID 4624 – Successful Logon

7. Investigation of Event ID 4625 – Failed Logon

Event ID 4625 represents a failed account logon attempt. Figure 2 shows the Security log filtered to Event ID 4625, which returned 237 matching events.

Observed Evidence

The Event Viewer list shows multiple entries marked “Audit Failure” under the Logon task category, with Source Microsoft Windows security auditing. The selected event shows Event ID 4625, Level Information, Logged 29-08-2026 2:47:09 PM, Computer WIN-BCDCP359AAS, and the general description “An account failed to log on.” The event list shows repeated 4625 entries across several dates, including entries on 29-08-2026, 04-07-2026, 30-06-2026, 10-06-2026 and 03-06-2026, indicating that failed logon attempts occurred on more than one occasion over time.

SOC Relevance

Failed authentication events are a key monitoring signal for a SOC analyst because a pattern of repeated 4625 events can indicate password guessing, brute-force attempts, or misconfigured services and scheduled tasks using outdated credentials. The available evidence shows that 237 failed logon events exist in the Security log; however, this screenshot alone does not establish the source, targeted account, or intent behind these failures, and it does not prove that the activity constitutes a

brute-force attack. Further correlation of the Account Name, Logon Type and source information (examined in Section 8) is required before drawing that conclusion.

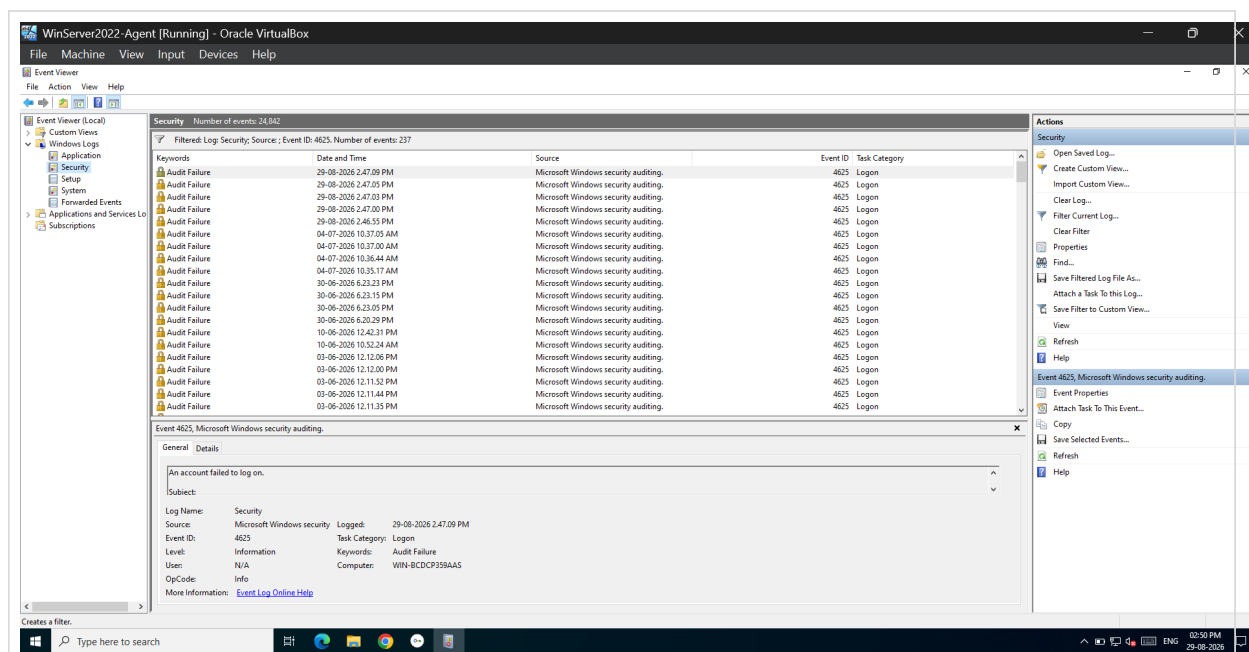


Figure 2: Windows Security Event ID 4625 – Failed Logon

8. Detailed Analysis of Event ID 4625

Figure 3 shows the Event Properties dialog for a selected Event ID 4625 entry, providing additional fields beyond the summary view in Section 7.

Observed Evidence

The Event Properties window displays the following visible fields for the selected failed logon event: Security ID: SYSTEM, Account Name: WIN-BCDCP359AA\$, Account Domain: WORKGROUP, Logon ID: 0x3E7, and Logon Type: 2. The General tab confirms Log Name: Security, Source: Microsoft Windows security, Event ID: 4625, Task Category: Logon, Level: Information, Keywords: Audit Failure, Logged: 29-08-2026 2:47:09 PM, and Computer: WIN-BCDCP359AAS.

SOC Relevance

These detailed fields allow a SOC analyst to investigate the source and context of a failed authentication event more precisely than the summary view alone. The Account Name and Account Domain identify which account the failed logon attempt was associated with, the Logon ID provides a unique session identifier that can be used to correlate related events, and the Logon Type indicates the method of logon being attempted (Logon Type 2 corresponds to an interactive logon). It should be noted that the Security ID shown in this particular event is SYSTEM and the Account Name shown is the machine account (WIN-BCDCP359AA\$) with Account Domain WORKGROUP; this combination is commonly associated with local system or service-level logon activity rather than an interactive end-user login, and the available screenshot does not include additional fields such as source workstation name or source network address that would normally be reviewed to fully characterize the failure.

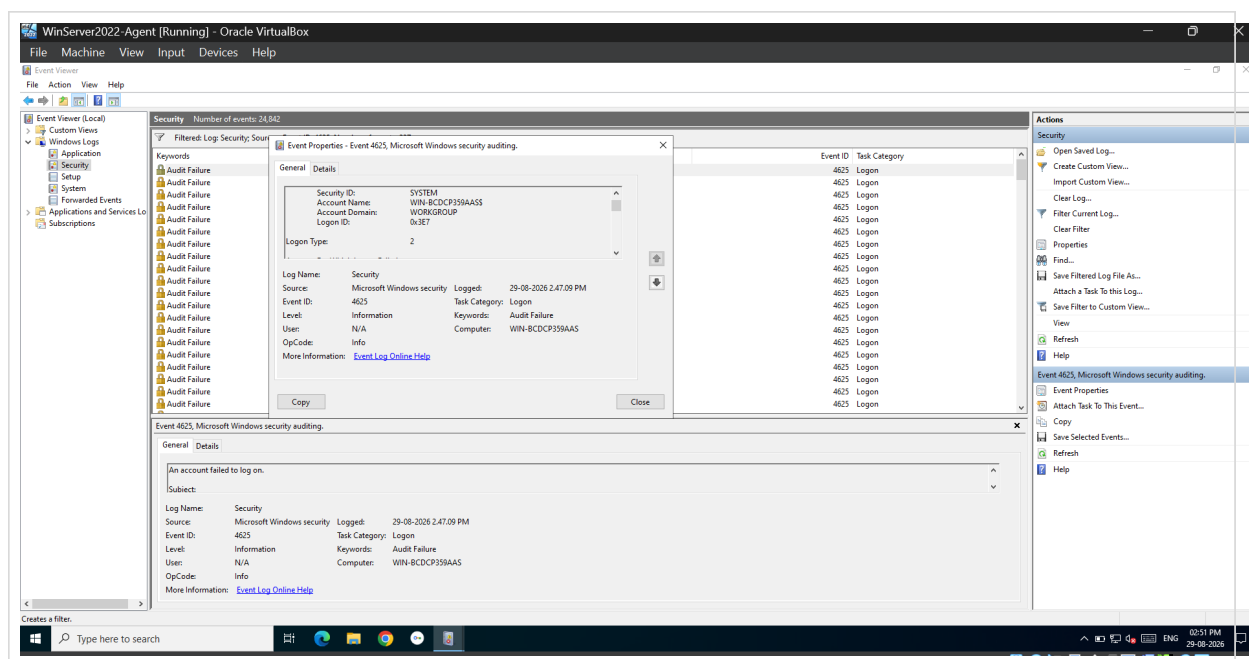


Figure 3: Event ID 4625 – Failed Logon Event Properties

9. Investigation of Event ID 4688 – Process Creation

Event ID 4688 records the creation of a new process on the host. Figure 4 shows the Security log filtered to Event ID 4688, which returned 300 matching events.

Observed Evidence

The Event Viewer list shows multiple entries marked “Audit Success” under the Task Category “Process Creation,” with Source Microsoft Windows security auditing. The selected event shows Event ID 4688, Level Information, Keywords Audit Success, Logged 29-08-2026 2:46:28 PM, Computer WIN-BDCP359AAS, and the general description “A new process has been created.” The Creator Subject field is visible in the description pane, though the specific account and process details within it are not fully readable in the screenshot provided.

SOC Relevance

Process-creation telemetry is a valuable source of information for SOC investigations because it allows an analyst to correlate process activity with other events, reconstruct execution chains, and identify processes that may be associated with suspicious or unauthorized behavior. In a full investigation, an analyst would review the Creator Subject (the account and logon session that started the process), the New Process Name, the Process Command Line (if command-line auditing is enabled), and the Parent Process, in order to determine whether the process activity is expected administrative behavior or something that warrants escalation. Because these specific field values are not clearly legible in the available screenshot, this report does not identify or characterize any individual process as suspicious.

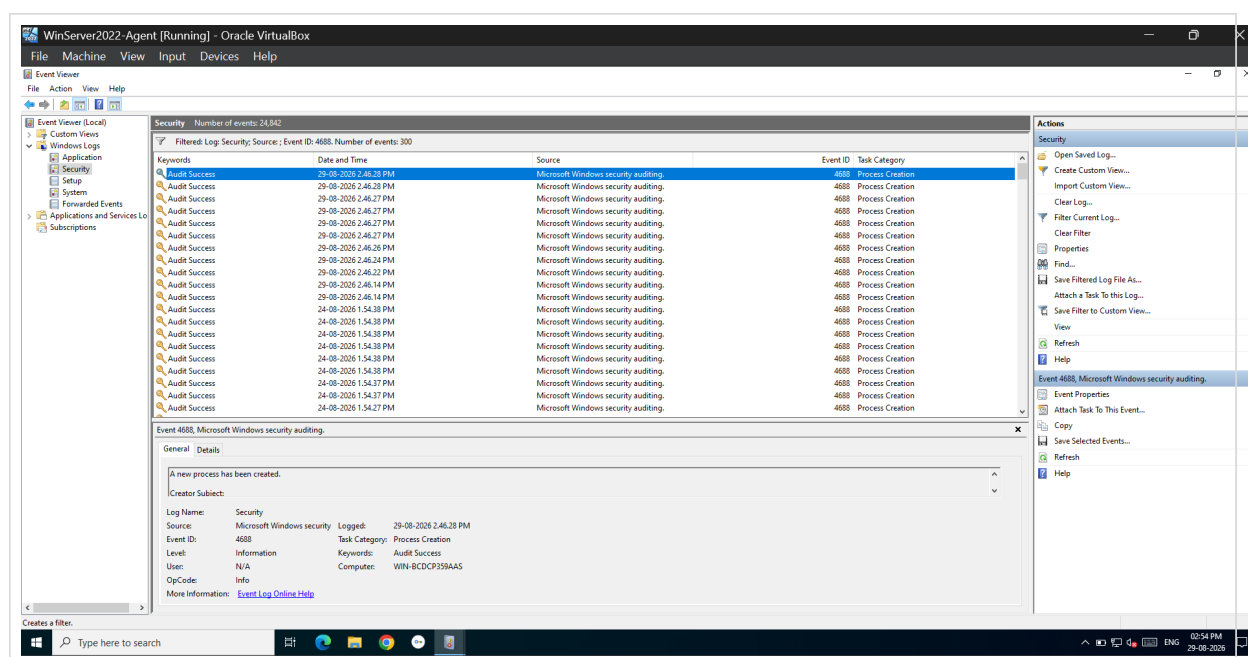


Figure 4: Windows Security Event ID 4688 – Process Creation

10. Investigation of Event ID 4720 – User Account Creation

Event ID 4720 records the creation of a new user account. Figure 5 shows the Security log filtered to Event ID 4720, which returned 4 matching events — notably fewer than the other three Event IDs reviewed.

Observed Evidence

The Event Viewer list shows four entries marked “Audit Success” under the Task Category “User Account Management,” dated 29-05-2026, 29-05-2026, 29-05-2026 and 28-05-2026. The selected event shows Event ID 4720, Level Information, Keywords Audit Success, Logged 29-05-2026 7:19:09 PM, Computer WIN-BCDCP359AAS, and the general description “A user account was created.” The Subject field is visible in the description pane but the specific account name creating the new user is not clearly legible in this screenshot.

SOC Relevance

Unexpected user account creation is an important event category for SOC monitoring because attackers who gain administrative access sometimes create new accounts to establish persistence or to maintain unauthorized access even if the original point of entry is later discovered and remediated. In this instance, four account-creation events were recorded on the host, which is a normal occurrence in many lab and administrative contexts. The available screenshot does not identify who created the accounts, what accounts were created, or the business justification for the creation, and therefore this report does not characterize these events as suspicious. This distinction — general SOC relevance of the event type versus what was actually confirmed from the evidence — is an important part of avoiding false conclusions during an investigation.

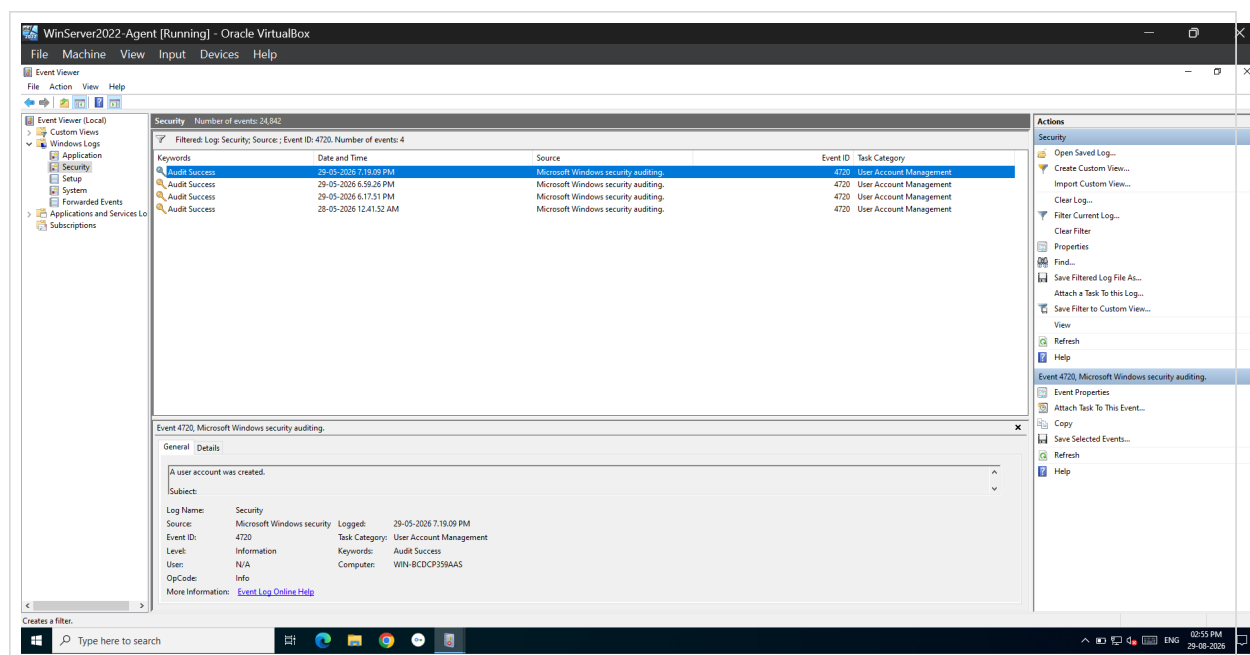


Figure 5: Windows Security Event ID 4720 – User Account Creation

11. TryHackMe Investigating Windows Activity

As part of Task 5, the TryHackMe room “Investigating Windows” was completed. This room presents a scenario in which a Windows machine has allegedly been compromised, and requires the learner to investigate the machine and answer questions based on artifacts found on the system.

Observed Evidence

Figure 6 shows the room in progress, with the banner indicating “Room completed: 100%” and Task 1 (“Investigating Windows”) questions answered, including “What’s the version and year of the windows machine?” (answered Windows Server 2016, marked Correct Answer) and “Which user logged in last?” (answered Administrator, marked Correct Answer). The Windows Settings panel visible in the same screenshot shows the target machine details: Edition Windows Server 2016 Datacenter, Version 1607, OS Build 14393.2791. Figure 7 shows the room summary page after completion, confirming “Room completed: 100%”, Task 1 marked as completed, and a notification that the lab machine was terminated successfully at the end of the session.

Relevance

This activity provided practical, hands-on exposure to Windows investigation and forensic analysis concepts, including identifying operating system version information and reviewing user logon history on a target system — skills that are directly applicable to the Event Viewer investigation performed earlier in this report.

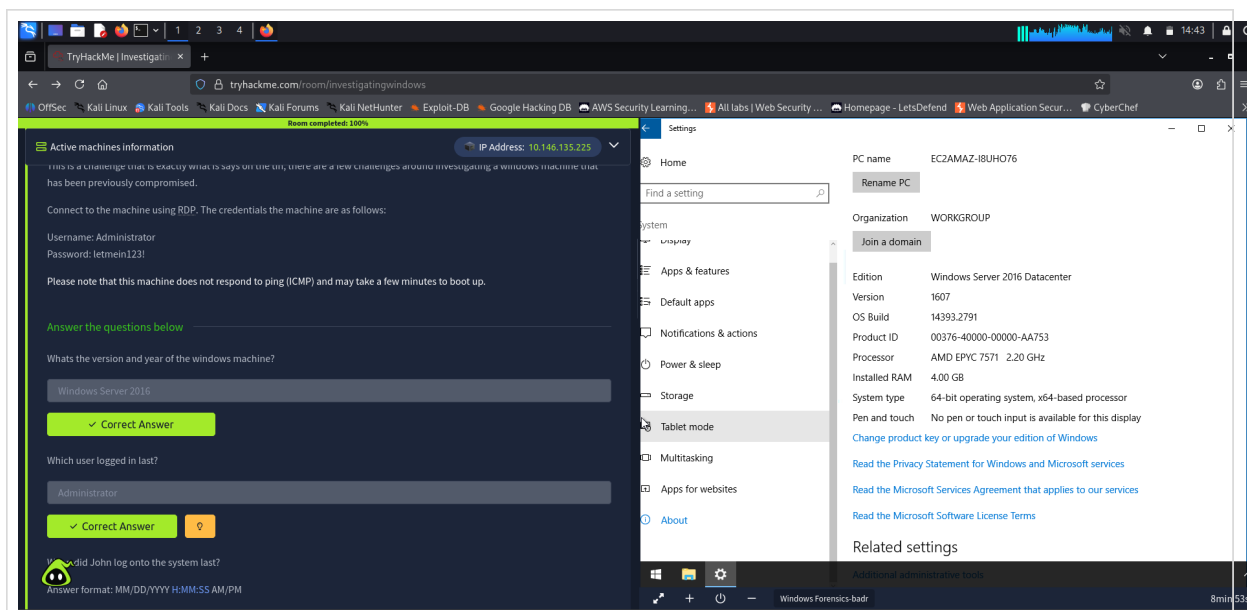


Figure 6: TryHackMe Investigating Windows – Task Completion

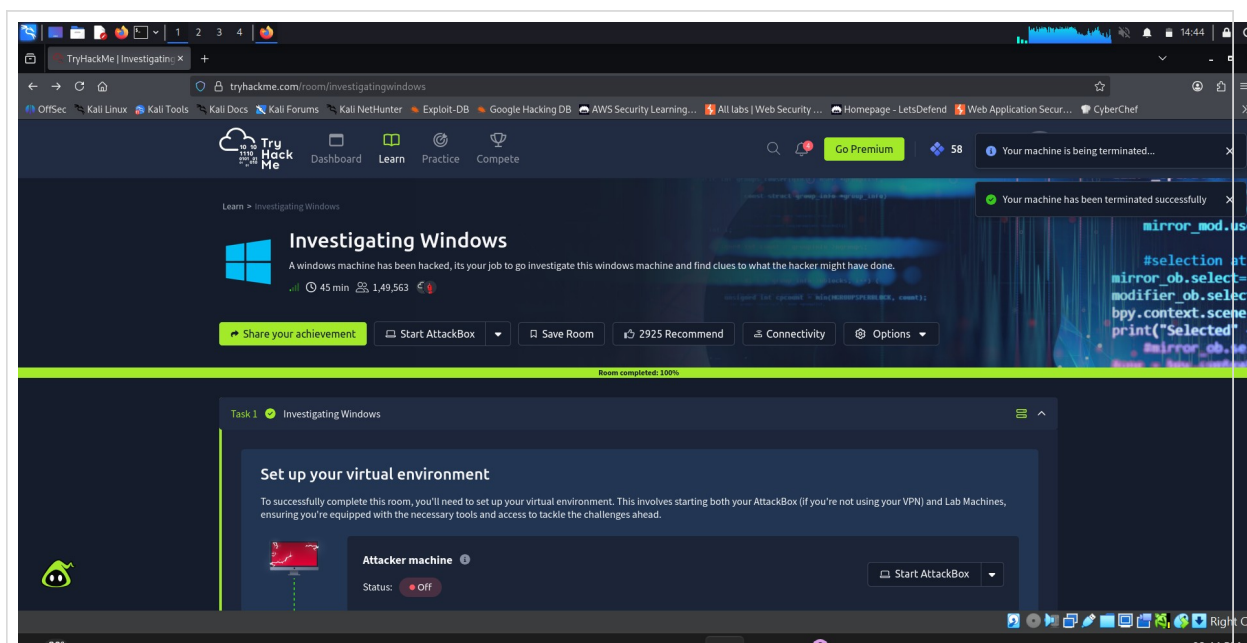


Figure 7: TryHackMe Investigating Windows – Room Completion

12. TryHackMe Windows Fundamentals 1 Activity

In addition to the investigation activity, the TryHackMe room “Windows Fundamentals 1” was completed. This room covers foundational Windows operating system concepts, including the Windows desktop, the NTFS file system, User Account Control (UAC), the Control Panel and related topics.

Observed Evidence

Figure 8 shows the room summary page with “Room completed: 100%” and the task list, including Task 1 (Windows Editions), Task 2 (The Desktop (GUI)), Task 3 (Introduction to Windows), Task 4 (The File System) and Task 5 (The Windows\System32 Folders), each marked with a completed checkmark.

Figure 9 shows the associated AttackBox environment connecting, with the room again showing “Room completed: 100%” and Tasks 1 through 3 marked complete, alongside target machine details including Hostname THM-WINFUN1 and Private IP Address 10.48.144.6.

Relevance

Completion of this room reinforced core knowledge of the Windows operating system, desktop environment and file-system concepts that underpin effective SOC log analysis, since understanding normal Windows behavior is a prerequisite for recognizing abnormal or suspicious activity in Security Event Logs.

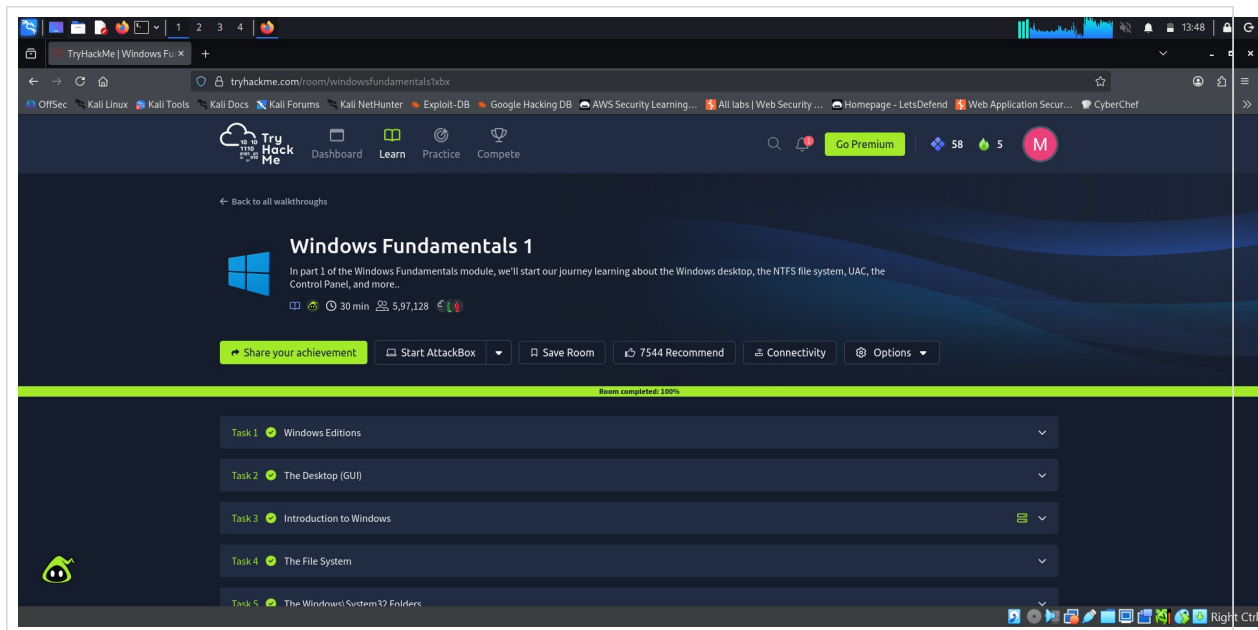


Figure 8: TryHackMe Windows Fundamentals 1 – Completion

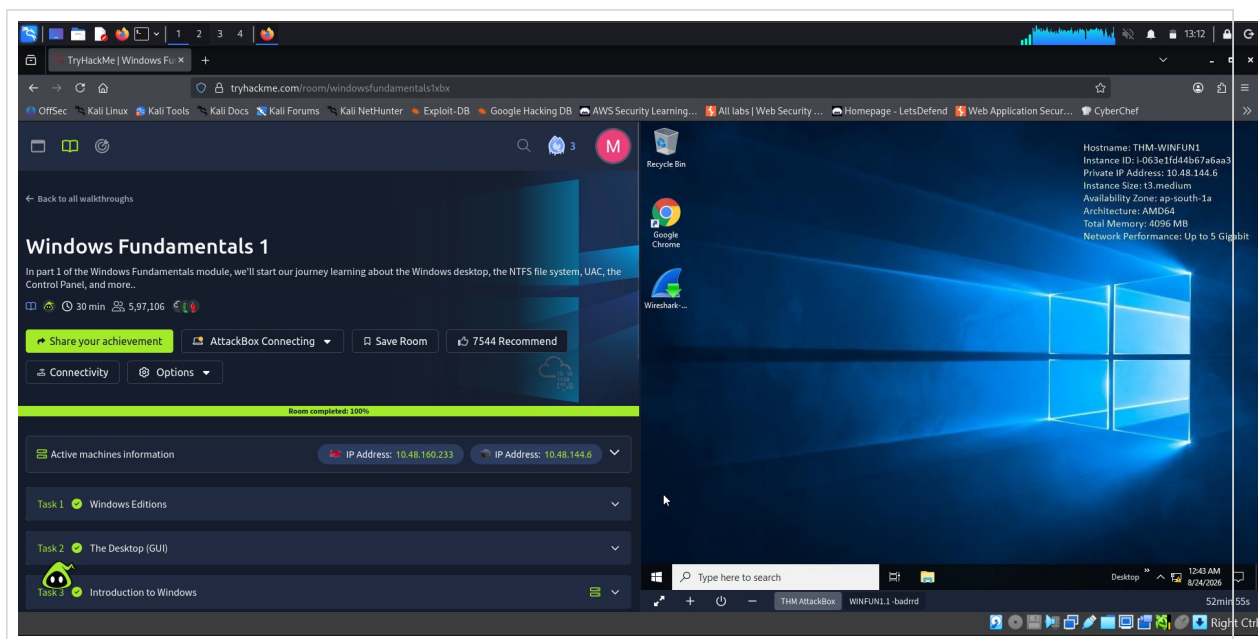


Figure 9: TryHackMe Windows Fundamentals 1 – Tasks Completed

13. SOC Analyst Investigation Methodology

The Event Viewer evidence reviewed in this task illustrates several core elements of SOC investigation methodology applied to Windows Security Event Logs:

- Monitor authentication activity by reviewing Event ID 4624 (successful logon) and Event ID 4625 (failed logon) to understand who is logging on, when, and how often.
- Identify repeated failed logons by tracking the volume and timing of Event ID 4625 entries, which can indicate password guessing or misconfigured systems.
- Investigate successful logons by reviewing Event ID 4624 details such as Logon Type and Account Name to confirm whether access was expected and legitimate.
- Review process creation activity through Event ID 4688 to understand what programs were executed on the host and by which account.
- Detect unexpected account creation by monitoring Event ID 4720 for new accounts that were not authorized or expected.
- Correlate multiple events (4624, 4625, 4688, 4720) using shared attributes such as Logon ID, Account Name and time proximity to build a fuller picture of activity on the host.
- Establish a timeline of activity by ordering relevant events chronologically to understand the sequence in which authentication, process execution and account changes occurred.
- Escalate potentially suspicious findings for further investigation when correlated evidence indicates activity that falls outside expected patterns, rather than acting on a single event in isolation.

14. Security Relevance and Detection Opportunities

Event Correlation: Event IDs 4624, 4625, 4688 and 4720 can be correlated during a security investigation to build context around a sequence of activity on a host. As a general, illustrative example of how correlation works — and not a claim that this occurred in this lab — a SOC analyst might treat a pattern of repeated Event ID 4625 failures followed by a successful Event ID 4624 logon on the same account as worth reviewing, particularly if it is then followed by unusual Event ID 4688 process creation or an unexpected Event ID 4720 account-creation event. Such a sequence would typically prompt further investigation into the source of the logon attempts, the Logon Type used, and the specific processes or accounts involved, rather than an immediate conclusion of compromise.

No such correlated sequence was confirmed from the screenshots reviewed for this task; the events examined were reviewed individually by Event ID filter rather than as a linked incident timeline.

Potential MITRE ATT&CK Context

The following MITRE ATT&CK concepts are noted only as general context that can reasonably be associated with the type of event examined, and are not presented as confirmed attacker behavior observed in this lab:

- Event ID 4625 (failed logon) can relate to techniques under Credential Access, such as Brute Force (T1110), when failures are repeated and systematic.
- Event ID 4688 (process creation) can relate to techniques under Execution, such as Command and Scripting Interpreter (T1059), when correlated with suspicious command-line content.

- Event ID 4720 (account creation) can relate to techniques under Persistence, such as Create Account (T1136), when an account is created without authorization.

These mappings represent general detection context associated with each Event ID category and are not confirmed findings from the evidence reviewed in this task.

15. Key Observations

Observed Evidence	Security Significance	Recommended Investigation
4,981 Event ID 4624 (successful logon) entries in the Security log.	Establishes a baseline of authentication activity on the host.	Review Logon Type and Account Name for logons occurring outside normal patterns.
237 Event ID 4625 (failed logon) entries, recurring across multiple dates.	May indicate authentication problems or attempted unauthorized access; not confirmed as an attack.	Correlate Account Name, Logon Type and timing to determine source and intent.
Detailed 4625 event showing Account Name WIN-BCDCP359AA\$, Domain WORKGROUP, Logon Type 2.	Identifies the account and logon type associated with a specific failure.	Review source workstation / network address fields (not visible in this evidence) for full context.
300 Event ID 4688 (process creation) entries.	Provides execution telemetry for the host.	Review Creator Subject and process name/command line for unusual or unauthorized execution.
4 Event ID 4720 (user account creation) entries.	New accounts were created on the host; general persistence risk if unauthorized.	Confirm the identity of the account creator and business justification for each new account.

No finding in this report should be read as confirmation that the reviewed host was compromised. The observations above describe what was visible in the supplied evidence and the standard SOC investigation steps that would follow, not a confirmed incident conclusion.

16. Skills and Knowledge Gained

- Practical use of Windows Event Viewer, including log filtering by Event ID and reviewing event properties.
- Security event analysis for authentication, process-creation and account-management activity.
- Failed logon (Event ID 4625) analysis, including interpretation of Account Name, Account Domain, Logon ID and Logon Type fields.
- Process-creation (Event ID 4688) monitoring concepts and their use in correlating execution activity.
- User-account activity monitoring (Event ID 4720) and its relevance to detecting unauthorized persistence.
- Foundational Windows forensic investigation skills, reinforced through the TryHackMe Investigating Windows room.

- Core Windows operating system fundamentals, reinforced through the TryHackMe Windows Fundamentals 1 room.
- SOC alert investigation methodology, including event correlation and timeline-based analysis.

17. Recommendations

- Monitor Event ID 4625 for repeated authentication failures against the same account or from the same source.
- Investigate unusual successful logons (Event ID 4624), particularly those with unexpected Logon Types or occurring outside normal business hours.
- Monitor Event ID 4688 for suspicious process execution, including unusual parent-child process relationships and command-line arguments where available.
- Monitor Event ID 4720 for unexpected user account creation and validate the identity of the account creator.
- Correlate authentication, process-creation and account-management events using shared identifiers such as Logon ID and Account Name.
- Maintain centralized log collection so that Security Event Logs from multiple hosts can be reviewed and correlated consistently.
- Use SIEM correlation and alerting to automatically flag patterns such as repeated failed logons followed by a successful logon.
- Investigate events based on user, time, host and logon context rather than in isolation.

18. Conclusion

This task provided practical exposure to Windows security-event monitoring, authentication analysis, process monitoring, account-management monitoring and basic SOC investigation methodology. Reviewing Event IDs 4624, 4625, 4688 and 4720 on the lab Windows Server host demonstrated how each event type contributes distinct telemetry to a SOC investigation, from confirming successful and failed authentication attempts to tracking process execution and account-management changes. The completion of the TryHackMe Investigating Windows and Windows Fundamentals 1 rooms further reinforced these concepts through hands-on practice in a guided lab environment.

Analyzing individual events in isolation provides only partial context; the greater value for a SOC analyst comes from correlating multiple event types across a consistent timeline to identify patterns that may warrant further investigation or escalation. This task strengthened foundational skills in Windows log analysis and SOC investigation methodology that are directly applicable to ongoing security-monitoring responsibilities within the internship.

19. Evidence and Screenshot References

Figure	Description	Source File
Figure 1	Windows Security Event ID 4624 – Successful Logon	01_event_viewer_event_id_4624.png
Figure 2	Windows Security Event ID 4625 – Failed Logon	02_event_viewer_event_id_4625.png

Figure 3	Event ID 4625 – Failed Logon Event Properties	03_event_id_4625_failed_logon_analysis.png
Figure 4	Windows Security Event ID 4688 – Process Creation	04_Event_Viewer_Event_ID_4688_Process_Creation.png
Figure 5	Windows Security Event ID 4720 – User Account Creation	05_Event_Viewer_Event_ID_4720_User_Account_Creation.png
Figure 6	TryHackMe Investigating Windows – Task Completion	06_TryHackMe_Investigating_Windows_Task_Completion.png
Figure 7	TryHackMe Investigating Windows – Room Completion	07_TryHackMe_Investigating_Windows_Completed.png
Figure 8	TryHackMe Windows Fundamentals 1 – Completion	08_TryHackMe_Windows_Fundamentals_1_Completed.png
Figure 9	TryHackMe Windows Fundamentals 1 – Tasks Completed	09_TryHackMe_Windows_Fundamentals_1_Tasks_Completed.png